Inverse Trace: A Ten-Positive Consistency Polynomial

Research note, 25 October 2026; prepared with AI assistance

1 An inverse correction composed with the explicit Pell trace

The supplied polynomial has ten independently strictly positive integer unknowns, exact total degree 232, and an evaluator with 85 live binary operations: 51 multiplications, 18 additions and 16 subtractions. Squares and fixed scalar multiplications are charged; all leaves are integers. The matched incoming *Trace Radical* graph has 87 operations at the same degree. This is an ordinary proof composition at the explicitly retained native/syntactic-refutation calibration, not a consistency proof, global minimum theorem, or new formal-kernel object.

1.1 Complete equation

The inputs are $J, F, \alpha, i, T, w, s, n, \eta, \zeta > 0$. The four fixed integers β, K_0, M_C, M_F are supplied as complete payloads and by a finite seed recipe. Every following symbol is a polynomial abbreviation, not an additional unknown:

$$
\begin{gathered}
h = \beta J, \qquad q = h + 1, \qquad C = q - F - \alpha, \\
X = qw, \qquad Y = qs, \qquad k = \eta + \zeta, \qquad c = kY + \eta, \\
A = Y(X + 1) + 1, \qquad \Delta = A^2 - 1, \qquad H = 4A - 5, \\
\ell = (XY)(kY), \qquad E = 1 + \ell(\ell + k), \\
R = [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\
B = (K_0 + w)C - F, \qquad U = ic^2, \qquad a = \Delta U^2 E, \\
F_0 = 1 + a, \qquad G = EF_0, \qquad f = a + F_0, \qquad Q = 4\Delta a F_0, \\
b = cT - R, \qquad V = fb - c, \qquad L = bV, \qquad K = X(3c - 2X) + 2, \\
e_6 = 6(k - R - 1) - \ell(k^2 - 4), \qquad D_6 = 6q^6, \qquad M = hD_6H, \\
Z = hD_6[H(n - LB) + K] - H[D_6B + h(e_6 - 6q^4 s)].
\end{gathered}
$$

The single ordinary integer polynomial equation is

$$\boxed{P = Q(Z^2 - M^2GV^2) + (M^2G - Z^2) = 0.}$$

There are no division gates, variable powers, binomial primitives or hidden quantified variables. For proof only put

$$v = XY^2, \quad e_* = k - R - 1 - \frac{vk(k^2 - 4)}{6}, \quad \rho = \frac{e_*}{q^6}, \quad \sigma = \frac{s}{q^2}, \quad \kappa = \frac{K}{H}, \quad t = \frac{B}{h}, \quad W = 1 + hL.$$

Before assuming any divisibility, the literal carrier identity is

$$z := Z/M = n + \kappa + \sigma - Wt - \rho. \tag{1}$$

1.2 Signed soundness and the corrected index

The inherited trace-linear root extraction uses

$$f^2 = 1 + 4aF_0, \quad Q = 4\Delta^2U^2G, \quad \gcd(E, F_0) = 1.$$

At a zero it yields $G = g^2$, $E = \tau^2$, $F_0 = f_0^2$, integral $z = \varepsilon gy$, and $QV^2 - (Q-1)y^2 = 1$. The pair $(f, I) = (1+2a, 2Ug)$ satisfies $f^2 - \Delta I^2 = 1$ and $c^2 \mid I$. Moreover $f \equiv 1 \pmod{c}$ identically. Signed folding gives $c = \psi_A(p)$ with positive odd p, and $k = 2\psi_{1+2v}(m)$ with $m \geq 2$. Thus $c > 8XY^3 \geq 8q^4$ and $K > 0$.

The retained main-index-three exclusion gives $c > c_3 = 4\Delta + 3$. The trace-linear auxiliary-index-three identities are

$$f(b - 4\Delta f) = c - c_3 \qquad \text{or} \qquad f(b + 4\Delta f) = c + c_3.$$

Their right sides are strictly between 0 and $2c < f$; hence the auxiliary index is at least five. The consequent bounds give $bV > 0$, $gy < L/2$ and $|b|/q^6 < L/2$. Because $k \geq 2$, the inverse correction is nonnegative before its subtraction: $e_* \leq k - R - 1 < b$, regardless of the sign of R. If $B \leq -1$, then $Wt < -L$ and (1) forces $n < L/2 - L + |b|/q^6 < 0$. Therefore $B \geq 0$ and $C > 0$. Only now use the calibrated mask estimate $3q < R < q^4 - 2$. The residues $V \equiv -c \pmod{f}$ and $V \equiv -R \pmod{c}$ give $p = R$; then R is odd and q is even.

The unbounded Lucas expansion is

$$k = 2m + 8vC_m + 32v^2 Z_m, \qquad C_m = \binom{m+1}{3}, \quad Z_m \geq 0.$$

Since k is even, $k(k^2 - 4)/6$ and e_* are integral. Reduce the literal Z modulo D_6, using $Z = Mz$; cancel the common ordinary integer factor 6, not a residue-ring unit. Since $\gcd(q, hH) = 1$, this proves

$$q^6 \mid e_* - q^4 s, \qquad q^4 \mid e_*.$$

For $t_0 = 2m$, $d = 2m - R - 1$, $A_m = C_m + 4vZ_m$ and $S_m = k^2 + kt_0 + t_0^2 - 4$, the exact cubic difference identity gives

$$3(e_* - d) = 4v^2(24Z_m - A_m S_m).$$

As $v = q^3 w s^2$, every prime of q gives $\operatorname{ord}_p(e_* - d) \geq 6 \operatorname{ord}_p(q) - \operatorname{ord}_p(3) \geq 4 \operatorname{ord}_p(q)$. Monotonicity yields $m < R$; consequently $|d| < R < q^4$ and $d = 0$. It is not yet legitimate to infer $q^6 \mid e_*$.

1.3 Project before recovering the full denominator

Reduction modulo H gives $H \mid 6hK$. Write $R = 2r + 1$. The authenticated ratio theorem gives $r \geq 48$, $Y > X^r/3$ and $X^{r+1} < H$. Universal projection yields

$$H \mid 6h(X - 2^R)(2^R X + 1).$$

With $u = 2^R = 2 \cdot 4^r$, the whole product satisfies

$$6h|(X - u)(uX + 1)| < X^4(X + u)(uX + 1) \leq 4 \cdot 4^r X^6 + 8 \cdot 16^r X^5 < X^{r+1} < H.$$

The last strict inequality follows at the minimum $(X, r) = (32, 48)$ from $2^{-117} + 2^{-25} < 1$, and improves thereafter. Thus $X = 2^R$, $H \mid K$, and q is a power of two. Now 3 is coprime to q, so the corrected-index identity proves $q^6 \mid e_*$. The carrier then gives $q^2 \mid s$ and $h \mid B$. Hence $q^3 \mid Y$ and $t \geq 0$ is integral, exactly the retained native reconstruction interface at source base $Y(X + 1) + 2$.

1.4 Strictly positive completeness

A finite refutation supplies fresh canonical outer data. Choose a positive Pell period seed with $c^2\tau \mid I_0$ and set $i = I_0/(c^2\tau)$. The inherited positive auxiliary pair can be kept while replacing its old input by

$$T = \frac{V + c + Rf}{cf} = fT_{\text{old}} - \frac{f - 1}{c} R > 0.$$

It is integral because $c \mid f - 1$ and satisfies $V = f(cT - R) - c$. The inverse correction can make ρ negative, but the explicit bounds

$$|\rho| < vk^3/3 < g/3, \qquad \kappa + \sigma < g/2$$

show that

$$n = gy + (1 + hL)t + \rho - \kappa - \sigma$$

is a strictly positive integer. The carrier is $z = gy$, proving $P = 0$. Both existential directions compose with the retained calibrated syntactic-refutation theorem; negation gives the equivalence with Con(ZFC).

1.5 Exact cost, coefficients and reproducibility

Deleting $v = (XY)Y$, sharing $\ell = (XY)(kY)$, and replacing the rank cubic by k^2, $k^2 - 4$, $\ell(k^2 - 4)$ saves two multiplications. For $q_0 = \beta J$, $k_0 = \eta + \zeta$ and $B_0 = k_0 sT - q_0^2(q_0 - F)$, the complete leading component is

$$[P]_{232} = -9 \cdot 2^{10} q_0^{100} w^{26} s^{58} i^{10} k_0^{32} B_0^2.$$

It is nonzero, not merely nonzero along a tested ray. The standard-library checker verifies 37 exact cuts covering all 85 gates, all 202 leading monomials, the full coefficient bytes and 171 rejected operator/missing-correction mutations. Normal and optimized receipts agree; finite recurrence diagnostics are separately labelled.

The principal tuple retains the incoming radix67 calibration. A second complete specialization uses the upstream one-anchor7309 tuple, credited rather than newly claimed: 172011 minimal raw bytes and largest port 345498 bits. Its complete native proof and finite Pc1 certificate are replayed with the new tuple; no old controller is substituted. The specialization is explicitly big-endian, unlike the principal little-endian payloads. All four ports are replaced together.

1.6 A small admissible-denominator interpolation

There is also a separate 77-operation,degree-348 graph in `q4_interpolation/`. It composes the retained q^4-denominator parent with the same unsquared trace; no new index correction is needed. Use the preceding definitions with $Y = q^3 s$, $D = q^4$, $e = k - R - 1$ and

$$M = hDH, \qquad Z = hD[H(n - bVB) + K] - H[DB + he].$$

The final polynomial is unchanged in form. Here $q^4 \mid e$ authenticates $2m = R + 1$; the full recurrence gives $4XY^2 \mid e$, and $q^4 \mid 4XY^2$ makes $\rho = e/q^4$ integral. The trace-linear signed bootstrap,projection and positive completeness then apply with $\sigma = 0$ and $\rho \geq 0$. The complete leading form is

$$[P]_{348} = -2^8 q_0^{216} w^{26} s^{60} i^{10} k_0^{34} T^2.$$

The independent checker verifies all 77 gates through 33 exact cuts and rejects 154 operator mutations. This is a composition/interpolation between the incoming 76/352 and 79/290 points,not a priority claim.

A Complete finite recipe for the principal coefficient tuple

In this appendix K denotes the port K_0 in the new equations. Every symbol introduced here is a fixed constant, independent of the ten quantified variables. The formulas and 436-character seed are the retained indexed BFS943 radix67 specification; they reconstruct the exact supplied payloads, with $\beta = 2^{590069} - 1$.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\Phi(x) = (1+2^{51})G_{943}(x) - 2^{10}\sum_{i=0}^{942} v(t_i)x^i + \sum_{(b,e,m)\in\mathcal{C}} 2^b x^e G_m(2x),$$

$$\mathcal{C} = \{(10,1023,8),(18,1013,9),(27,1003,8),(35,983,9),(44,1031,1),(45,1022,1),(46,1011,1),(47,992,1),(48,1095,1),(49,2195,1),(50,3296,1)\}.$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$h(q,b) = 2(q+1)+b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\},$$
$$\mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\},$$
$$r_q = 6 + |\mathcal{R} \cap \{0,\dots,q-1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0,\dots,q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w,n,e,s)$
$0, 1$	$(2,435,3,1),\ (5,0,0,0)$
$j+2\ (j = 2q+b < 430)$	$((1-d_j)l_{u_j}, h(q,b), d_j r_{u_j}, v_j)$
$432+b\ (b=0,1)$	$(0, h(215,b), 0, h(215,b))$
$434+b\ (b=0,1)$	$(0,b,0,b)$
$424+2r_q+b\ (q\in\mathcal{R}, b=0,1)$	$(r_q, b, 0, h(q,b))$
$424+2l_q+b\ (q\in\mathcal{L}, b=0,1)$	$(0, b, l_q, h(q,b))$
$924+j\ (0 \le j < 4)$	$(0, j+430\lfloor j/2\rfloor, 0, 438)$
928	$(2,434,2,0)$
929	$(3,436,4,4)$
930	$(4,436,4,0)$
$497+t\ (t = 434,435,436)$	$(1,438,1,t)$
$934+3a+b-6\lfloor a/4\rfloor$ $a\in\{0,1,4\},\ b=0,1,2$	$(a,437,b,437)$

For $t = (w,n,e,s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\,2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A,k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b\rfloor$. For $0 \le q < 215$, $b = 0,1$, $j = 2q+b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \lfloor \mathcal{A}/2^{636} \rfloor \bmod 2^{1675},$$
$$E_0 = \lfloor \mathcal{A}/2^{2311} \rfloor,$$
$$\nu_j = D_2(N,j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430+q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1-m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1+\sigma_j) + (1-\nu_j)D_{216}(T_0, j-\sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1,1,1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0,\dots,61$, is

`0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`.

Concatenate the following 436 characters as digits $e_0, \dots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

B Complete finite recipe for the one-anchor7309 tuple

Here K denotes the fixed port K_0, not an unknown. The entire tuple is from the credited upstream one-anchor7309 construction and uses minimal unsigned big-endian payloads. The source mask shift occurs exactly once.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{7309}, \quad \beta = B - 1,$$
$$G_m(x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{4161} + \rho^{5228}.$$

$$\begin{aligned} M_C &= \beta + \rho - (\rho+3)\rho^{1063} \\ &\quad - G_{943}(\rho) - \Psi(\rho, 1), \\ M_F &= B + 3 + (2^{46} - 2)\Lambda \\ &\quad + \rho^{1013} G_9(\rho) + \rho^{1065} G_8(\rho) + \rho^{2079} \\ &\quad + 2^{25}\rho^{6343} - \rho^{6325}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1137} G_{943}(\rho) + \rho^{7351} \\ &\quad + \Lambda\Phi(1/\rho), \end{aligned}$$

where

$$\begin{aligned} \Phi(x) &= \sum_{i=0}^{942} [1 + 2^{10}(1 + w_i + 2^8 n_i \\ &\quad + 2^{17} p_i + 2^{25} s_i)] x^i + 2^{10}\Psi(x, 2), \\ \Psi(x, t) &= x^{1003}(x^{20} + t^{17}) G_8(tx) \\ &\quad + t^8 x^{983}(x^{30} + t^{17}) G_9(tx) \\ &\quad + t^{34}(2t - 1) x^{2079}. \end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2$ $(j = 2q + b < 430)$	$((1 - d_j) r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$
$434 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{424 + 2r_{d,q} + b \\ q \in \mathcal{S}_d,\ d,b = 0,1}$	$(dr_{d,q}, b, (1 - d) r_{d,q}, h(q, b))$
$924 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
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$497 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$934 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A,k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \lfloor \mathcal{A}/2^{636} \rfloor \bmod 2^{1675},$$
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$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(E_0, \tau_q) - 14),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j)D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

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0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```